MEMO

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Subject: Leak detection simulator (compensated volume balance): summary of results
Code and full results: github.com/snur1981-dev/leak-detection-sim

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Purpose

I built this as a self-study project to learn how computational pipeline monitoring (CPM) works. With no CPM work experience, I wrote a compensated volume balance in Python and tested it against the four performance measures in API RP 1130: sensitivity, reliability, accuracy and robustness. All data is synthetic, generated by my own simplified model of a made-up pipeline. The results show how the method behaves under my assumptions; they are not a validation of a real system.

Setup

- **Pipeline:** 100 km, NPS 16 (390.6 mm ID), flat, crude oil at 850 kg/m³, normal flow 0.20 m³/s (720 m³/h). Flow meters at both ends, pressure at 0, 50 and 100 km, temperature at both ends.
- **Hand calculations:** friction loss 6.10 MPa at full flow, line pack 11.52 m³ per MPa (about 74% oil compression, 26% pipe stretch) and 9.59 m³ per °C, wave speed 1,106 m/s.
- **Test data:** 1-minute SCADA-style readings with meter bias and noise, leaks, pump starts, valve moves and data faults. True values are kept separately for scoring; the detector only reads the tags.
- **Detector:** imbalance = flow in − flow out − change in line pack, over 5 minute, 1 hour and 24 hour windows. An alarm needs 3 minutes in a row over the threshold. Alarms are held for 15 minutes after a pump or valve status change, and data quality checks run before the balance.
- **Thresholds:** a bias allowance (0.1% of flow, assumed after meter proving) plus 3 times the noise. The noise was measured from a 7 day no-leak tuning run, the way an operator would tune from historian data.

Results

1. Line pack correction removes pump-start false alarms. A pump start from 0.10 to 0.20 m³/s raises average pressure by about 2.2 MPa and packs about 25 m³ into the line. The plain balance (flow in minus flow out) counted that as missing oil, reached 25.2 m³ and alarmed at 06:04. The compensated balance stayed under 0.7 m³. Both caught the real 1% leak at 10:28 (Figure 1, using a 3 m³ placeholder threshold).

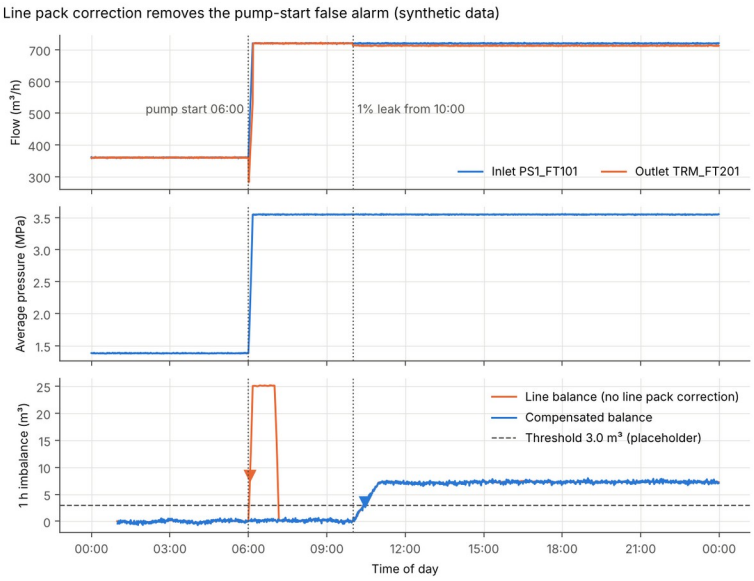


Figure 1. One simulated day with a pump start at 06:00 and a 1% leak at 10:00 (synthetic data).

2. Thresholds and sensitivity. With the tuned thresholds, the 1 hour window caught a 1% leak in a median of 8 minutes and caught every 0.5% leak. The 24 hour window was slower on these sizes because it already holds a full day of meter bias, so a leak has to add more volume to cross its threshold. Its job is leaks below about 0.2%, which my test matrix did not include.

	5 min window	1 h window	24 h window
Threshold (m^3)	0.67	1.44	18.93
Minimum detectable leak, design estimate	1.11%	0.20%	0.11%
Time to alarm, 0.5% leak (median, 5 seeds)	2 of 5 caught	18 min	89 min
Time to alarm, 1% leak	11 min	8 min	47 min
Time to alarm, 2% leak	4 min	5 min	25 min
Time to alarm, 5% leak	3 min	3 min	10 min

3. Meter bias sets the floor. Noise averages out over longer windows. Bias does not, because it adds the same volume every minute, just like a leak. Most of the 24 hour threshold (17.3 of 18.9 m^3) is the bias allowance, and the smallest detectable leak levels off near 0.1% of flow. Bias also made every leak rate estimate about 0.6 m^3/h too high (a 20% error on a 0.5% leak, 2% on a 5% leak). Only better meters or regular proving lower this floor.

4. Reliability. Over 7 days with no leak, 6 pump stop and start pairs and 2 valve moves (5 seeds), the plain balance averaged 1.14 false alarms per day in the 5 minute and 1 hour windows, one each time flow increased. The compensated balance had none. The 15 minute inhibit made no difference here, because my model has no pressure waves, so compensation left nothing for it to catch.

5. Robustness (clock skew and deadband). An outlet clock running a few seconds late (up to 10 s) caused no false alarms. False alarms started around 30 s, once the added imbalance approached the 1 hour threshold. A 0.5 m^3/h reporting deadband on the flow meters caused none, but 2 m^3/h, larger than the meter noise, caused 76 false alarms in the 1 hour window over 21 days: stuck values act like a temporary bias (Figure 2).

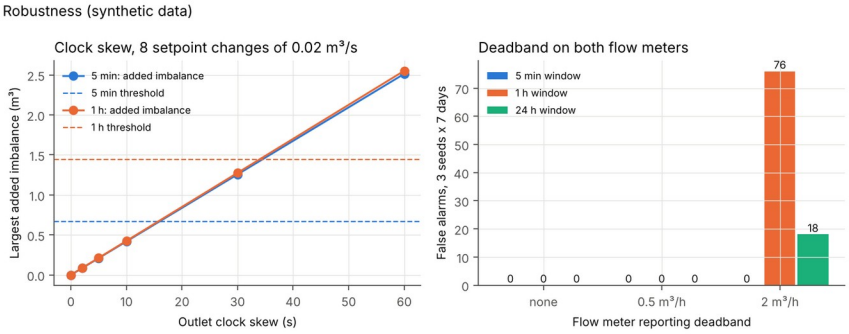


Figure 2. Effect of outlet clock skew (left) and flow meter reporting deadband (right) on false alarms (synthetic data).

6. Data quality checks. The checks caught a frozen outlet reading and stopped its false alarms, but did not catch slow meter drift (0.02% per day), which looks the same as a growing leak. During a 30 minute outlet dropout, suppressing the bad data delayed the alarm on a real 2% leak (82 minutes instead of 25 in the 1 hour window). A CPM should tell the controller that detection is degraded rather than go quiet.

Limitations

What the model simplifies: flat line with no elevation or slack line; one crude with fixed properties and no batches; each minute treated as settled, so no pressure waves (about 90 s to cross the line); meter errors as constant bias plus clean random noise; the leak does not change the pressures; a 7 day tuning run is short.

What a real system also has to handle: batch tracking, slack line and shut-in conditions, meter proving and maintenance, SCADA communications and time sync across many sites, alarm management and controller response procedures, and leak location.

Next steps: test leaks below 0.2% of flow to show what the 24 hour window adds, add pressure waves and elevation and retest the inhibit, and run the method on real historian data with a senior advisor.

Notes: written in Python (numpy, pandas, matplotlib) with help from an AI tutor. I did the hand calculations, ran every test and checked the results against them. I studied the standards from public summaries of API RP 1130, API TR 1149 and CSA Z662 Annex E. All numbers come from results/summary.md in the repository.